House Price Data Cleaning & Preprocessing

Complete data cleaning pipeline for construction data: villas, apartments, Homecity, Towncity, and Happy Homes

Made with GAMMA

Project Overview

Complete Pipeline

20+ data cleaning techniques applied to transform messy data into ML-ready dataset

Real-World Data

Synthetic house price data simulating construction sector with realistic data issues

ML-Ready Output

Cleaned dataset prepared for machine learning applications

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Technologies & Features

Core Technologies

- Python 🐍
- Pandas
- NumPy
- Scikit-learn

Dataset Features

- City location
- House type
- Built year
- Price & area
- Bedrooms & bathrooms
- Garage availability

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Data Generation & Initial Cleaning

01

Data Generation

Created synthetic dataset with 1,000 rows

02

Missing Values

Introduced 5% null values randomly

03

Handle Missing

Price → median, Area → mean, City → "Unknown"

04

Remove Duplicates

Eliminated duplicate records

05

Fix Data Types

Ensured proper data type consistency

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Text & Data Quality Cleaning

Remove Whitespace

Trimmed extra spaces from text fields

Correct Typos

Fixed "Housston" → "Houston"

Remove Outliers

Applied IQR method for Price

Normalize Data

Applied MinMaxScaler on Price & Area

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Feature Engineering & Encoding

Encode Categorical

Label encoding for City & House_Type

Feature Engineering

Created House_Age from Built_Year

Date Features

Extracted Listing_Year

Column Renaming

Area_sqft → Area_in_SqFt Built_Year → Year_Built

Column Management

Dropped Agent_Name Reordered columns

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Final Validation Steps

- **Handle Zero Values**

  Checked for invalid zero entries

- **Remove Special Characters**

  Cleaned non-printable characters

- **Handle Negative Values**

  Removed invalid negative values

- **Detect Imbalance**

  Checked City distribution

- **Final Validation**

  Verified unique values

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EDA & Analysis Results

1K

Dataset Rows

Initial synthetic dataset size

10+

Features Analyzed

Complete EDA completed

Exploratory Data Analysis completed with visualizations for all features

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Machine Learning Implementation

ML Code

Written based on EDA analysis

Model Training

Applied to cleaned dataset

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Streamlit Application

Web Interface

Built interactive application using Streamlit framework

Deployment

Complete ML pipeline integrated into web app

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